

ZhiGui

Decision & Planning Assistant · Second Brain Skill

Version 1.0.0

Node.js ≥ 17

Windows / macOS / Linux

2026-07-31

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CHAPTER 1

Product Overview

ZhiGui is a conversation-triggered personal assistant system composed of three core components:

1.1 Assistant Positioning

ZhiGui is not a simple to-do list tool. Its design is inspired by three classic role models, each corresponding to a core capability layer:

Secretary

Helps you see the big picture, schedule your day, and prepare morning briefings and reviews. At the start of every conversation, it already knows what you have planned today, which tasks are overdue, and which goals are in progress.

Butler

Quietly manages all daily affairs — note archiving, topic classification, action tracking, and decision recording. You never need to worry about how data is stored or where; the butler takes care of it for you.

Mentor

Inspired by the Mentor character in classic fantasy literature — a teacher hidden deep in your mind. It won't make decisions for you, but when you need it, it offers precise advice: what you shouldn't do, which goals are fantasies, which decisions will sow hidden risks.

Second Brain

Underlying all three roles is a single knowledge graph: notes, goals, decisions, and schedules are interlinked through foreign keys, forming a traceable memory network.

Role Switching

ZhiGui never initiates conversations on its own — it is not a background daemon. When you ask it, it reports like a secretary; when you instruct it, it executes like a butler; when you hesitate, it reminds you like a mentor. All three roles are driven by the same dataset, and switching is seamless.

1.2 System Components

ZhiGui consists of three core parts:

MCP Engine

A backend JSON-file-driven data layer that exposes 40+ tool calls to AI assistants via the Model Context Protocol, with no need to run independently.

Electron Desktop Panel

A visualization panel that docks to the screen edge, supports collapse/expand toggling, and lets you directly edit schedules, goals, notes, and to-do items.

AI Skill Protocol

An operational specification defined in SKILL.md. The AI loads a Bootstrap package at the start of every conversation to obtain the full index of your context.

Second Brain

Notes, goals, decisions, and schedules are interlinked through foreign keys like topicId / noteId / goalId, forming a traceable knowledge relationship graph.

Data Synchronization

Panel operations and AI conversations share the same data entities, synchronized in real time through file watching (fs.watch). If you edit a schedule in the panel, the next AI conversation sees it immediately; notes created by the AI are auto-refreshed in the panel.

1.3 System Architecture

```
User ↔ AI Assistant (Trae/Cursor, etc.)
    ↳ MCP Protocol
ZhiGui Engine (server.js)
    ↳ JSON File Read/Write
.zhigui/ Data Directory
    ↳ fs.watch File Watching
Electron Panel (main.js)
    ↳ IPC Communication
Frontend UI (dashboard.js)
```

Installation

2.1 Prerequisites

- **Node.js ≥ 17** : Required to run the MCP engine and Electron panel. Download at <https://nodejs.org>
- **Git** (optional): For cloning the project from GitHub
- **An MCP-capable AI tool**: Such as Trae, Cursor, etc.

2.2 Option 1: Download ZIP for Direct Use (Recommended)

1 Download and Extract the Project

Download the ZIP from GitHub: <https://github.com/CarlWangChina/zhigui-openclaw-ui-second-brain-skill>

Extract to any directory, for example `D:\ZhiGui`. The extracted directory structure:

```
D:\ZhiGui\  
├─ start.bat           ← Windows one-click launch script  
├─ start.sh           ← macOS/Linux launch script  
├─ skill\  
│   ├─ engine\  
│   │   └─ dashboard\  
│   │       └─ electron\  
│   │           └─ lib\  
│   │               └─ scripts\  
│   │                   └─ test\  
│   │                       └─ SKILL.md  
│   │                           └─ config.json  
│   │                               └─ mcp-config-template.json  
│   │                                   └─ package.json  
│   └─ package.json  
└─ README.md
```

← MCP engine + business logic
← Web panel (server.js + public/)
← Electron desktop shell (main.js + preload.js)
← Configuration & data initialization
← Install & seed scripts
← 15 test suites
← AI skill protocol document
← Engine configuration
← MCP config template
← Dependency declaration
← Electron dependencies

2

First Launch (Auto-installs Dependencies)

Double-click `start.bat` (Windows) or run `./start.sh` (macOS/Linux).

The script will automatically perform the following operations:

- Detect and install Node.js dependencies (`npm install`)
- Download the Electron binary
- Run `setup.js` to initialize the data directory and configuration
- Register the global MCP server
- Launch the Electron desktop panel

First-run Notice

The first launch downloads Electron (about 100MB); please be patient. An npmmirror mirror is preconfigured to accelerate downloads for users in mainland China.

3

Verify Successful Installation

When a narrow panel window appears on the right side of your desktop with the title bar showing "ZhiGui", the installation is successful.

If you want to load demo data to see it in action, run the following in the project's `skill\` directory:

```
node scripts/seed-demo-data.js
```

This creates 3 sample notes, 2 goals, 1 decision, and a 3-day schedule to help you quickly understand the system's features.

2.3 Option 2: Manually Install the Skill into an AI Tool

If your AI tool supports a "skill directory" (such as Trae's `~/.trae-cn/skills/`), you can copy the contents of the `skill\` directory there:

```
# Using Trae as an example
copy /E /I skill\ %USERPROFILE%\trae-cn\skills\zhigui\
```

Or use the project's built-in setup script to do it automatically:

```
node skill/scripts/setup.js <project_directory> <skill_directory> <node_path>
```

2.4 Option 3: Run the Web Panel Only (Without Electron)

If you don't want to install Electron, you can run just the web server:

```
cd skill
node dashboard/server.js
```

Then open `http://localhost:7788` in your browser to use the panel.

CHAPTER 3

Configuring MCP to Connect AI Assistants

MCP (Model Context Protocol) is the bridge through which AI assistants invoke ZhiGui tools. Once configured, the AI can read and write your schedules, notes, goals, and other data.

3.1 Configure in Trae

1 Open the MCP Configuration Page

In Trae: Settings → MCP → Add → Manual Add

2 Paste the MCP Configuration

Paste the following JSON, replacing `<skill_directory>` with your actual path:

```
{
  "mcpServers": {
    "zhigui": {
      "command": "node",
      "args": ["D:/ZhiGui/skill/engine/server.js"]
    }
  }
}
```

```
}  
  
}  
  
}
```

3 Save and Verify

After saving, Trae will automatically connect to the MCP server. In a conversation, say "Help me check today's schedule". If the AI calls the `zhigui_get_assistant_bootstrap` tool and returns data, the configuration is successful.

3.2 Configure in Other AI Tools

Any MCP-capable AI tool (Cursor, VS Code Copilot, etc.) can use the same configuration. Refer to the `mcp-config-template.json` template file in the project root directory.

Configuration Tips

- If `node` is not in PATH, use the full path: `"C:/Program Files/nodejs/node.exe"`
- Paths in JSON should use forward slashes `/` to avoid backslash escaping issues
- No env variables need to be set — the engine auto-detects the data directory
- The web panel (port 7788) does not need to be running for MCP to work
- The MCP server reads and writes JSON files directly; the panel auto-refreshes via fs.watch

3.3 Configuration File Reference

`config.json` is located in the skill directory and controls where the engine points its data directory:

Field	Description	Example
<code>dataDir</code>	Data storage directory (self-contained relative path, or absolute path)	<code>.zhigui</code>
<code>appDir</code>	App root directory (<code>.</code> in standalone mode)	<code>.</code>

CHAPTER 4

UI Panel Feature Guide

The panel has two forms: **expanded mode** (full-feature panel, docked to the right edge of the screen) and **collapsed mode** (a small icon in the bottom-right corner; click to restore expanded mode).

4.1 Top Toolbar

Toolbar Buttons (left to right)

- **Logo / Subtitle**: Displays the "ZhiGui" brand name and the "Decision & Planning Assistant" subtitle
- **Language Toggle**: Switch between Chinese and English (affects all UI text)
- **Theme Toggle**: Dark / light theme
- **Always-on-top Toggle**: Pin window always-on-top (state persisted to state.json)
- **Collapse Button**: Collapse the panel into a small bottom-right icon
- **Close Button**: Exit the application

4.2 View Tabs

The panel provides three view tabs:

Today

Morning briefing, schedule, to-do items, completed tasks, daily reflection — everything you need to focus on today.

Knowledge

Topic library, note index, full-text search — the knowledge indexing layer of your second brain.

All

Goals, constraints, decisions, notes — the complete list of all long-term entities.

4.3 Today View Feature Blocks

Today

Knowledge

All

TODAY AT A GLANCE

Jul 31

Demo | Outline project presentation mainline

4 pending • 0 completed.

4

pending

0

completed

0%

day progress

Deadline 2

+ Add to schedule

Add an action

● Today's guidance

Jul 31

Must Complete

Finish the project presentation mainline.; Confirm family follow-up appointment.

Today's Pick

This afternoon, verify cross-topic note and schedule links.

Not Recommended

Do not interrupt morning focus time just to add minor details.

Figure 1: Today view — overview, morning briefing, schedule, and to-do items

The large text area at the top shows today's date, total task count, in-progress goal count, today's completion percentage, and attention signal badges (e.g., "2 overdue · 1 blocked").

Two quick-action buttons are provided: **Add to Schedule** (opens the event-add dialog) and **Add Item** (opens the to-do-add dialog).

Morning Briefing Area (Decision Check)

A decision briefing generated by the AI during your first conversation of the day, containing 5 fixed blocks:

Block	Marker Color	Description
Must Do	Red	Tasks that must be completed today; leaving them undone has consequences
Recommended Today	Green	Items the AI recommends handling today
Not Recommended	White	Things to avoid doing today and why
Strategic Reminder	Gold	Directional reminders for long-term goals
Daily Quote	Gold	An inspiring daily phrase

The morning briefing is a "today-only" feature — the engine refuses to write a briefing for any date other than today. If you view a future day, the panel shows a "Preview (not today's briefing)" label.

Schedule Area

Displays all task cards for the day, organized by date. Each task card shows:

- Time, duration, title, description
- Linked goal, linked topic, linked notes (click to expand)
- Completion checkbox (click to toggle completion status)
- Right-click menu: Edit time, Delete task

The top of the schedule area has date navigation: < previous day, > next day, **calendar icon** opens a month picker to select a date.

To-do Items Area (Errands)

A queue of to-do items that have not been assigned a date. Each card shows the title, priority (must/should/nice), and linked notes. Click to edit or complete.

Completed Area (Completed Today)

Today's completed to-do items, shown with strikethrough and semi-transparent styling. Each item has a **Restore** button to undo the completion status.

Daily Reflection Area

A review report generated by the AI during your last conversation of the day, including: a summary of completed items, goal health check, attention changes, and lifecycle management suggestions.

4.4 Knowledge View

Today

Knowledge

All

● Topic Library

All categories (3) ▾

Search notes, goals, topics...

Search

Work Notes 1 topics • 3 notes

▶ **Project Delivery**

Active

3 • 1

Health Records 1 topics • 2 notes

▶ **Family Health**

Active

2 • 0

Decision Principles 1 topics • 1 notes

▶ **Personal Planning**

Active

1 • 1

Last updated: 2026-07-31 16:20

Figure 2: Knowledge view — topic list, category filter, and full-text search

All topics are displayed as cards, showing note count, linked goal count, and linked action count. Supports filtering by category and full-text search.

The search bar supports full-text retrieval across notes, goals, and topics. Each topic can be expanded to view all note titles under it.

Notes Area

Notes are displayed as index cards (title and metadata only; body content is not preloaded). Click a card to expand and view the full content.

A "Record Note" button lets you manually enter notes directly in the panel. For file imports, attach files in chat and let the AI handle them.

4.5 All View

Today

Knowledge

All

● Goals



▶ Demo | Balance reliable delivery with family support

战略

🔒 1

● Constraints



No constraints
Tap + on top-right to add

● Decision Records

Accepted

Demo | Finish presentation mainline first, then schedule flexible comms

End tracking

The project demo needs continuous focus; family follow-up is better confirmed in...

Related goals: Demo | Balance reliable delivery with family support • Demo | Complete project

Figure 3: All view — goals, constraints, decision records, and notes

Displays strategic goals and current goals. Each goal card can be expanded to view linked tasks, linked notes, and decision records.

Goal Field Reference

- **why**: Why this goal matters
- **obstacle**: Current obstacle
- **risk**: Potential risk
- **confidence**: Confidence value
- **successCriteria**: Success criteria
- **nextStep**: Next action
- **deadline**: Deadline

Constraints Area

Long-term constraint conditions, such as "Sleep before 11 PM to protect eyesight". Constraints influence the AI's scheduling suggestions.

Decisions Area

Structured decision records, including title, rationale, impact, and linked entities. Decisions have 5 states: Pending, Accepted, Rejected, Expired, Superseded.

4.6 Collapsed Mode

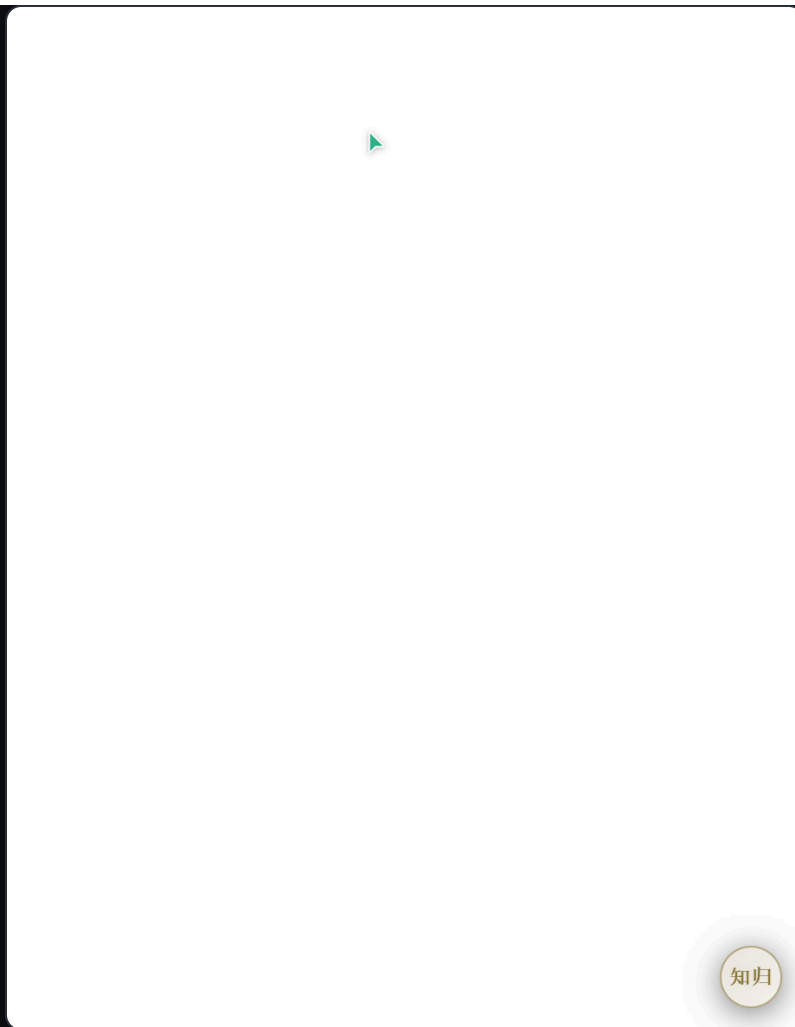


Figure 4: Collapsed mode — a circular "ZhiGui" icon in the bottom-right corner; click to restore

After clicking the collapse button, the panel shrinks into a small circular icon in the bottom-right corner (showing "ZhiGui"). Click the icon to restore the expanded panel. Window position and always-on-top state are persisted.

CHAPTER 5

Core Intelligent Features

What sets ZhiGui apart from ordinary to-do tools is a four-layer intelligent design: proactive linking, layered indexing, long-term memory, and a customizable skill protocol.

5.1 Proactive Linking of Notes

ZhiGui's core design philosophy: **connecting entities is the core value of a second brain, not an optional feature**. You don't need to manually tell the AI which note relates to which goal — the AI proactively searches and establishes the links.

How Proactive Linking Works

- When creating a task: the AI auto-scans the note index, finds relevant notes, and attaches them via `noteIds[]`
- When creating a goal: the AI searches the note library and links notes that support that goal
- When recording a note: after saving, the AI checks whether it supports an existing goal or to-do, and auto-backlinks
- When auto-scheduling: the engine automatically appends the goal's linked notes to derived tasks

Why This Matters

An unlinked note is a lost note. ZhiGui's proactive linking discipline ensures every piece of information is attached to a node in the knowledge graph — forming a traceable relationship network through foreign keys like `topicId`, `noteId`s, and `goalId`. The "Linked goal · 3 notes · 1 decision" you see in the panel is how this works.

5.2 Layered Indexing and On-demand Loading (Token Saving)

ZhiGui adopts a three-layer data architecture: **index first, details on demand**. This is the key to how it remembers all your information without blowing up the context window.

Layer	Name	Content	When Loaded
Layer 0	Compact Index	Only ID, title, category — enough to locate, not enough to fabricate details	Auto-loaded at the start of every conversation (Bootstrap package)
Layer 1	Full Details	The complete content of a single entity — goal description, note body, schedule	Loaded on demand only when the user asks about that entity
Layer 2	Precomputed Summary	Morning briefing, attention summary — conclusions the engine has already computed for you	First conversation of the day / when decision advice is needed

Token Saving Mechanism

The Bootstrap package (Layer 0) typically takes only 2–4K tokens and contains the index of all your notes, goals, and schedules. The AI decides from the index which entities need details, then calls tools like `zhigui_get_note_detail` to load individual Layer 1 records. This means even with thousands of notes, each conversation only consumes tokens relevant to the current topic.

5.3 Long-term Memory

ZhiGui's memory is **persistent file storage**, not session-level context. Your data exists as JSON files on your local disk, preserved permanently across sessions and tools.

Cross-session Memory

At the start of every new conversation, the AI calls `zhigui_get_assistant_bootstrap` to load the full index. Yesterday's notes, last week's goals, last month's decisions — all in memory.

Cross-tool Memory

Data lives in local JSON files, not bound to any AI platform. Use Trae today, switch to Cursor tomorrow, VS Code the day after — the same dataset, the same memory.

Activity Log

Every create, modify, and complete operation is recorded in the activity journal. The AI uses `zhigui_reconcile_activity` to catch up on panel changes made while you were offline, so nothing is missed.

Version Tracking

Each data change increments `stateVersion`. The AI uses it to determine what's new since the last conversation, loading only the delta rather than the full set.

5.4 Customize Your Butler: Editing SKILL.md

SKILL.md is ZhiGui's "soul file" — it defines the AI assistant's behavioral norms, role positioning, and operating procedures. By editing it, you can shape a butler of your own.

Core Customizable Parts of SKILL.md

- **Role positioning:** The default is a "secretary + butler + mentor" trinity. You can change it to a pure fitness coach, study advisor, writing assistant — just rewrite the opening `description` and role paragraphs
- **Retrieval rules:** Control when the AI loads details vs. uses only the index. Loosen them and the AI preloads more aggressively; tighten them to save tokens
- **Proactive linking discipline:** Defines how aggressively the AI searches and links notes when creating entities. Can be set to "link only when the user mentions it" or "auto-link via full search"
- **Morning briefing format:** The labels and content direction of the 5 fixed blocks (Must Do / Recommended / Not Recommended / Strategic Reminder / Daily Quote) can be customized
- **Reflection logic:** The report structure and focus dimensions (goal health, attention changes, etc.) of the daily reflection can be adjusted
- **Decision tree:** Controls when the AI proactively creates tasks vs. only suggests without executing

Example: Turn ZhiGui into a Study Advisor

```
# Open skill/SKILL.md and modify the opening section:
```

```
description: A study advising assistant. Use this skill when the user discusses study plans, review progress, subject scheduling, or time allocation.
```

```
# ZhiGui Study Advisor Protocol
```

```
You are a study advisor. Role positioning:
```

- Secretary: Manage daily review schedules and progress tracking
- Butler: Archive notes by subject, flag weak knowledge points
- Mentor: Remind you of goals when you slack off, dissuade you when you try to take on too much

```
## Morning Briefing Customization
```

- Must Do: Subjects and chapters that must be reviewed today
- Recommended Today: Review content suggested by the forgetting curve
- Not Recommended: Do not start more than 3 new subjects at once
- Strategic Reminder: XX days until the exam

Safety Notice

Editing SKILL.md does not affect data files; it only changes the AI's behavior. If something breaks, just re-download the original SKILL.md from GitHub and overwrite it. Back up the original file before editing.

CHAPTER 6

Usage Examples

6.1 Demo Data Overview

After running `node scripts/seed-demo-data.js`, the system creates the following sample data:

Demo Notes (3)

- **Family Health** topic: "Demo | Family follow-up appointment talking points" — Confirm the follow-up appointment time and transportation by phone this week
- **Project Delivery** topic: "Demo | Key info for project presentation" — Explain user value first, then demonstrate the end-to-end flow
- **Personal Planning** topic: "Demo | Scheduling principle: buffer for important relationships" — When work is compressible, prioritize keeping family time

Demo Goals (2)

- **Strategic Goal**: "Demo | Balance reliable delivery with family support" — Complete key work milestones while preserving time to care for family
- **Current Goal**: "Demo | Complete project presentation prep this week" — Due in 2 days; finish the presentation mainline today

Demo Schedule (3 days)

- **Yesterday**: 16:30 "Demo | Confirm presentation scope" (completed, used for reflection review)

- **Today:** 09:30 Demo | Outline project presentation mainline / 14:30 Demo | Verify cross-topic traceability / 20:00 Demo | Call to confirm family appointment
- **Tomorrow:** 10:00 Demo | Final check before sending demo materials

Demo Decision & Reminder

- **Decision:** "Demo | Finish presentation mainline first, then schedule flexible comms" (accepted)
- **Reminder:** "Demo | Check tomorrow morning if demo materials were sent" (tomorrow 09:00)
- **Morning Briefing:** Must do = Finish presentation mainline + Confirm family appointment; Recommended = Verify cross-topic notes afternoon; Not recommended = Don't interrupt morning focus time; Strategic = Work delivery and family support need clear time boundaries

6.2 Typical Conversation Scenarios

Scenario 1: Daily Scheduling

User: Help me plan today's study

AI (automated flow):

1. Calls `zhigui_get_assistant_bootstrap` to get context
2. Finds today's existing schedule: 09:30 Outline project presentation mainline
3. Finds constraint: Sleep before 11 PM to protect eyesight
4. Reads linked notes and goals
5. Suggests scheduling English study from 14:00 to 16:00
6. Reminder: You already have a high-focus task this morning; afternoon study is recommended

User: OK, add it

AI: Created task "English study" 14:00-16:00

Linked note "English" and goal "Exam English prep"

Scenario 2: Recording a Note and Linking It

User: I just called my family and confirmed the follow-up appointment – next Wednesday morning

AI:

1. Creates note: "Family follow-up appointment confirmed"
Topic: Family Health
Content: Follow-up appointment next Wednesday morning;
need to arrange transportation in advance
2. Auto-links to goal "Balance reliable delivery with family support"
3. Auto-links to existing note "Family follow-up appointment talking points"
4. Prompts: Do you want to add a reminder for next Wednesday in the schedule?

Scenario 3: Completing a To-do and Reflection

User: I've finished outlining the presentation mainline

AI:

1. Marks task "Outline project presentation mainline" as completed
2. Updates goal progress (1/3 completed today)
3. Checks what's next: tomorrow has "Final check before sending demo materials"
4. Generates a completion record in completedActions

(At the end of the day)

AI calls zhigui_get_reflection to generate a reflection:

- Completed 3 tasks today
- Goal "Project presentation prep" is on track
- Suggests prioritizing sending the materials tomorrow morning

Scenario 4: Panel Operations and AI Sync

User in the panel:

1. Clicks the "Add to Schedule" button
2. Fills in: tomorrow 15:00 "Team weekly meeting"

3. Saves

In the next conversation:

AI calls bootstrap and auto-discovers the new "Team weekly meeting"

AI: I noticed you have a team weekly meeting tomorrow at 15:00.

It doesn't conflict with "Send demo materials".

I suggest doing the final check at 16:00 after the meeting.

6.3 Panel Manual Operation Guide

Operation	Method	Description
Add schedule	Click "Add to Schedule" button	Fill in date, time, title, duration
Complete task	Click task card checkbox	Instantly toggles completion status; supports undo
Edit time	Right-click task card → Edit time	Opens time editor
Delete task	Right-click task card → Delete	Requires confirmation; references auto-cleaned after deletion
Add goal	All view → + button	Choose strategic goal or current goal
Add to-do	Click "Add Item" button	Fill in title, priority, links
Restore completion	Completed area → Restore button	Undo an accidental completion
View note	Click note card to expand	Body loaded on demand, not preloaded
Search knowledge	Knowledge view search bar	Full-text search across notes, goals, topics
Switch date	Schedule area < / > buttons	Or click the calendar icon to pick a date

FAQ

Q: What if the Electron download fails on first launch?

Run manually: `npm install --registry https://registry.npmmirror.com`, then `node node_modules/electron/install.js`. If it still fails, check your network proxy settings.

Q: The AI conversation reports an MCP connection failure?

Check that the path in the MCP configuration is correct. Run `node skill/engine/server.js` in a terminal to manually test whether the engine starts normally. If node is not in PATH, use the full path.

Q: The AI doesn't sync after panel operations?

The panel and AI share the same JSON files and synchronize through file watching. The AI reloads data via bootstrap at the start of every conversation. If the data seems inconsistent, just start a new conversation.

Q: How do I reset all data?

Delete all JSON files in the `skill/.zhigui/` directory, then re-run `node skill/lib/reset-data.js` or `node skill/scripts/seed-demo-data.js`.

Q: Why can I only view today's morning briefing?

ZhiGui's morning briefing is designed as a "decision record for the current day" — it's a decision document the AI writes for today's plan, not a generic schedule preview. To view a future day, use the `zhigui_get_day_schedule` tool to read a preview.

Q: The window position is wrong after collapsing?

In collapsed mode the window sits in the bottom-right corner of the screen (16px above the taskbar). In expanded mode it restores to full height on the right edge of the screen. After dragging the window, closing and reopening it will automatically save the position.

Q: What if topics are duplicated?

The engine has added normalize matching (trim + case-insensitive) to prevent duplicates. If duplicates still occur, ask the AI in a conversation to merge them using `zhigui_merge_topics`. The AI checks the existing topic list before creating an entity.

Q: How do I run the tests?

Run `npm test` in the `skill/` directory. It executes 15 test suites covering core features, the storage layer, the scheduler, the attention engine, and other modules.

ZhiGui · Decision & Planning Assistant · User Manual v1.1

GitHub: github.com/CarlWangChina/zhigui-openclaw-ui-second-brain-skill

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